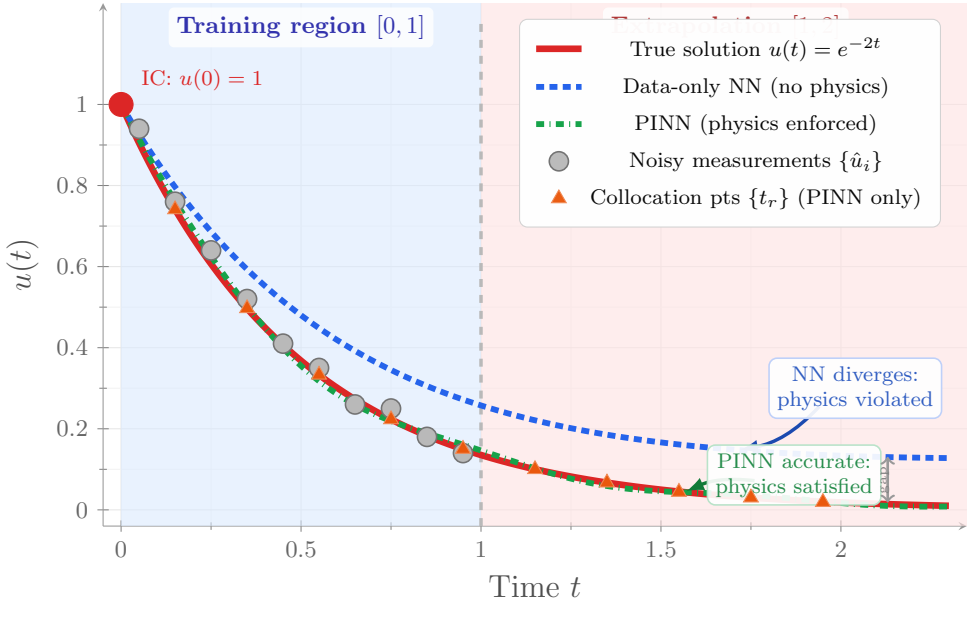
Training region $[0, 1]$
IC: $u(0) = 1$
True solution $u(t) = e^{-2t}$
Data-only NN (no physics)
PINN (physics enforced)
Noisy measurements $\{\hat{u}_i\}$
Collocation pts $\{t_r\}$ (PINN only)
NN diverges: physics violated
PINN accurate: physics satisfied
$u(t)$
Time t
0
0.2
0.4
0.6
0.8
1
0
0.5
1
1.5
2